Feasible normals and admissibility in the restricted bivariate normal problem

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Setup and statement

Let

$$X \sim N_2(\mu, I_2), \qquad \Lambda = \{\mu : \mu_2 \le \mu_1\} \ ,$$
$$\Lambda_0 = \{\mu \in \Lambda : \mu_1 \le 0\} \ , \qquad \Lambda_1 = \{\mu \in \Lambda : \mu_1 > 0\} \ .$$

A nonrandomized test is written as

$$\phi_A(x) = I\{x \notin A\} \ ,$$

where $A \subseteq \mathbf{R}^2$ is the acceptance region. Throughout this note, A is assumed to be closed, convex, and coordinatewise lower:

$$x \in A, \quad y \le x \quad \Longrightarrow \quad y \in A \ .$$

For a closed convex set C, write

$$h_C(a) := \sup_{x \in C} a'x$$

for its support function, allowing the value $+\infty$. Define the feasible normal cone

$$\mathcal{F} := \{a \in \mathbf{R}^2 : 0 \le a_2 \le a_1\} = \text{cone}\{(1,0),(1,1)\} \ .$$

The phrase "all finite normals lie in $\mathcal{F}$" means

$$a \ne 0, \quad h_A(a) < \infty \quad \Longrightarrow \quad a \in \mathcal{F} \ . \tag{1}$$

For a lower set, finite support already forces $a \ge 0$, so failure of (1) is equivalent to the existence of some $a \ne 0$ with

$$h_A(a) < \infty \ , \qquad 0 \le a_1 < a_2 \ .$$

The main result is the following characterization.

Theorem 1 (Feasible-normal characterization)**.** *Let $A \subseteq \mathbf{R}^2$ be closed, convex, and coordinatewise lower, and suppose*

$$P_0(X \in A) = 1 - \alpha \ , \qquad 0 < \alpha < 1 \ .$$

Then the test $\phi_A = I\{X \notin A\}$ is α-admissible, against all level-α tests, if and only if all finite normals of A lie in $\mathcal{F}$, that is, if and only if (1) *holds.*

The proof has two parts. The sufficient direction is the Lehmann–Romano ray argument. The necessary direction is constructive: if A has an X_2-heavy finite normal, then one can transfer null probability from an infeasible face to a feasible diagonal face and strictly increase power at every $\mu \in \Lambda_1$.

1. Sufficiency

We use the following standard consequence of the Lehmann–Romano complete-class result for exponential families.

Proposition 1 (Lehmann–Romano ray criterion)**.** *Consider an exponential family with natural statistic T and natural parameter space Ω. Let Ω_H be the null parameter set and let Ω' be an alternative set. Let A_0 be a closed convex acceptance region for a nonrandomized test. Suppose that whenever*

$$A_0 \subseteq \{t : a't \le c\}$$

for some nonzero a, there exist $\theta^* \in \Omega$ and $\lambda_n \to \infty$ such that

$$\theta^* + \lambda_n a \in \Omega' \ .$$

Then the test with acceptance region A_0 is d-admissible. If, in addition, the test has exact size α attained at a finite null point, then it is α-admissible. These conclusions allow randomized competing tests.

Lemma 1 (Sufficiency of feasible normals)**.** *If A is closed, convex, coordinatewise lower, satisfies $P_0(A) = 1 - \alpha$, and satisfies* (1)*, then ϕ_A is α-admissible against all level-α tests.*

Proof. Since $\Sigma = I_2$, the normal experiment is the canonical exponential family

$$\frac{dP_\mu}{dP_0}(x) = \exp\{\mu' x - \tfrac{1}{2}\mu'\mu\} \ ,$$

with natural parameter μ and statistic $T(x) = x$. The alternative set is

$$\Omega' = \Lambda_1 = \{\mu : \mu_1 > 0,\ \mu_2 \le \mu_1\} \ .$$

We verify the ray condition in Proposition 1. Suppose

$$A \subseteq \{x : a'x \le c\}$$

for some nonzero a. Then $h_A(a) < \infty$, so by (1), $a \in \mathcal{F}$. Since $a \neq 0$ and $0 \le a_2 \le a_1$, necessarily $a_1 > 0$. Hence, for every $\lambda > 0$,

$$\lambda a \in \Lambda_1 \ .$$

Thus the ray condition holds with $\theta^* = 0$ and any sequence $\lambda_n \to \infty$.

Because A is coordinatewise lower, the rejection region A^c is coordinatewise upper. For every $\mu \in \Lambda_0$, we have $\mu \le 0$. Coupling $X = \mu + Z$ with $Z \sim N_2(0, I_2)$ gives $\mu + Z \le Z$ coordinatewise, and therefore

$$P_\mu(X \notin A) \le P_0(Z \notin A) = \alpha \ .$$

Thus the test has level α, and exact size is attained at the finite null point $\mu = 0$. Proposition 1 gives α-admissibility against all level-α tests, including randomized competitors. □

2. The transfer order

The necessity proof uses the order with respect to which every restricted-alternative likelihood ratio is increasing. Define

$$T_1(x) := x_1 + x_2 \ , \qquad T_2(x) := -x_2 \ ,$$

and write

$$x \succeq_\Lambda y \quad \Longleftrightarrow \quad T_1(x) \geq T_1(y) \quad \text{and} \quad T_2(x) \geq T_2(y) \ .$$

The positive cone of this order is

$$\Pi := \{d : x + d \succeq_\Lambda x\} = \text{cone}\{(1,0),(1,-1)\} = \{d : d_2 \leq 0,\ d_1 + d_2 \geq 0\} \ .$$

Intuitively, Π is the cone of universally power-improving moves in sample space for the restricted alternatives. That is, the direction $(1,0)$ is a power-improving move because it increases x_1 directly, and the direction $(1,-1)$ is a power-improving move because it moves evidence from coordinate 2 to coordinate 1 while keeping $x_1 + x_2$ fixed.

For $\mu \in \Lambda_1$, write

$$u := \mu_1 > 0 \ , \qquad v := \mu_1 - \mu_2 \geq 0, \qquad \mu = (u, u - v) \ .$$

Then, relative to P_0,

$$L_\mu(x) := \frac{dP_\mu}{dP_0}(x) = \exp\left\{uT_1(x) + vT_2(x) - \frac{1}{2}(\mu_1^2 + \mu_2^2)\right\} \ . \tag{2}$$

Thus L_μ is strictly increasing in T_1 and weakly increasing in T_2.

We first replace A by its closed $-\Pi$-hull,

$$\widehat{A} := \text{cl}(A - \Pi).$$

Equivalently, $\widehat{A}$ is the smallest closed set containing A that is closed under moves opposite to the restricted likelihood-ratio order. This operation removes boundary pieces whose supporting normals point in infeasible directions. It may add too much acceptance probability, so we then restore the original size by cutting $\widehat{A}$ with a feasible diagonal halfspace.

For a rectangle, this hull operation keeps the x_1-gate but replaces the bad vertical face $x_2 = c_2$ by a diagonal face $x_1 + x_2 =$ constant. The next lemmas show that the same geometric repair works for any closed convex coordinatewise lower acceptance region.

Lemma 2 (The feasible-normal hull)**.** *The set $\widehat{A}$ is closed, convex, coordinatewise lower, and Π-lower, meaning*

$$x \in \widehat{A} \ , \quad p \in \Pi \quad \Longrightarrow \quad x - p \in \widehat{A} \ .$$

Moreover, every nonzero finite-support direction of $\widehat{A}$ lies in $\mathcal{F}$. If A has a finite-support direction a with $a_2 > a_1$, then

$$A \subsetneq \widehat{A} \qquad \textit{and} \qquad P_0(\widehat{A}) > P_0(A) \ .$$

Proof. Because $\widehat{A}$ is the closure of the Minkowski sum $A + (-\Pi)$, it is closed and convex. It contains A because $0 \in \Pi$. It is coordinatewise lower because A is coordinatewise lower: if $x = y - p$ with $y \in A$ and $p \in \Pi$, and if $z \le x$, then

$$z = (y - (x - z)) - p \ ,$$

where $y - (x - z) \in A$. Passing to the closure preserves the lower-set property. The Π-lower property is immediate from the definition.

The support function satisfies

$$h_{\widehat{A}}(a) = h_A(a) + h_{-\Pi}(a) \ ,$$

with the usual convention for extended-real sums. Now

$$h_{-\Pi}(a) < \infty \iff a \cdot (-1, 0) \le 0 \ \text{ and } \ a \cdot (-1, 1) \le 0 \iff a_1 \ge 0 \ \text{ and } \ a_2 \le a_1 \ .$$

Also, since A is coordinatewise lower and nonempty, $h_A(a) < \infty$ implies $a \ge 0$. Therefore, if $h_{\widehat{A}}(a) < \infty$ and $a \ne 0$, then

$$0 \le a_2 \le a_1 \ ,$$

so $a \in \mathcal{F}$.

Finally suppose that $h_A(a) < \infty$ for some a with $a_2 > a_1$. Then $h_{\widehat{A}}(a) = +\infty$, because a is not in the polar cone of $-\Pi$. Indeed, since $(-1, 1) \in -\Pi$ and $a'(-1, 1) = a_2 - a_1 > 0$, every point $x \in A$ generates the ray $x + t(-1, 1) \in A - \Pi$. Along this ray $a'x$ tends to $+\infty$. Hence the support of $\widehat{A}$ in direction a is infinite.

Hence $\widehat{A}$ is not contained in the supporting halfspace $\{x : a'x \le h_A(a)\}$ that contains A, so $\widehat{A} \setminus A$ is nonempty. Since $P_0(A) \in (0, 1)$ and A is convex, A has nonempty interior. Taking a point of $\widehat{A}$ outside that supporting halfspace and joining it to an interior point of A gives points of $\operatorname{int}(\widehat{A}) \setminus A$. Hence $\widehat{A} \setminus A$ contains a nonempty open set, and the Gaussian density is everywhere positive, so

$$P_0(\widehat{A}) > P_0(A) \ .$$

□

Lemma 3 (Calibration of the competitor)**.** *Assume $P_0(A) = 1 - \alpha$ and that A has a finite-support direction a with $a_2 > a_1$. Then there exists a finite number b such that*

$$P_0\big(\widehat{A} \cap \{x : T_1(x) \le b\}\big) = 1 - \alpha \ .$$

For any such b, define

$$A' := \widehat{A} \cap \{x : T_1(x) \le b\} \ .$$

Then A' is closed, convex, coordinatewise lower, has exact null probability $P_0(A') = 1 - \alpha$, and all its finite-support directions lie in $\mathcal{F}$. Consequently, $\phi_{A'}$ has level α, attained at $\mu = 0$.

Proof. By Lemma 2, $P_0(\widehat{A}) > P_0(A) = 1 - \alpha$. The map

$$t \mapsto P_0\big(\widehat{A} \cap \{T_1 \le t\}\big)$$

is continuous, tends to 0 as $t \to -\infty$, and tends to $P_0(\widehat{A})$ as $t \to \infty$. Hence at least one finite b gives value $1 - \alpha$.

The set A' is an intersection of two closed convex coordinatewise lower sets, so it is closed, convex, and coordinatewise lower. Its exact null probability is built into the choice of b. To check the finite-support directions, note that

$$\mathrm{rec}(A') = \mathrm{rec}(\widehat{A}) \cap \{d : T_1(d) \le 0\} \ .$$

Thus the polar cone of $\mathrm{rec}(A')$ is contained in the closed convex cone generated by the finite-support directions of $\widehat{A}$ and the vector $(1, 1)$. Lemma 2 puts the former directions in $\mathcal{F}$, and $(1, 1) \in \mathcal{F}$. Since $\mathcal{F}$ is a closed convex cone, every finite-support direction of A' lies in $\mathcal{F}$.

Finally, because A' is coordinatewise lower, the same coupling argument used in Lemma 1 gives

$$\sup_{\mu \in \Lambda_0} P_\mu(X \notin A') = P_0(X \notin A') = \alpha \ .$$

Thus $\phi_{A'}$ has level α, attained at 0. □

Lemma 4 (Transfer domination criterion)**.** *Let G and D be Borel sets with*

$$P_0(G) = P_0(D) > 0 \ .$$

Suppose that, for every $g \in G$ and $d \in D$,

$$T_1(g) > T_1(d) \qquad \textit{and} \qquad T_2(g) \ge T_2(d) \ .$$

Then

$$P_\mu(G) > P_\mu(D) \qquad \textit{for every } \mu \in \Lambda_1 \ .$$

Proof. Let $m = P_0(G) = P_0(D)$. Using the likelihood ratio (2),

$$m\{P_\mu(G) - P_\mu(D)\} = \int_G \int_D \{L_\mu(g) - L_\mu(d)\} \, dP_0(d) \, dP_0(g) \ .$$

For $\mu \in \Lambda_1$, the coefficient on T_1 in (2) is strictly positive and the coefficient on T_2 is nonnegative. Therefore the displayed ordering gives

$$L_\mu(g) > L_\mu(d)$$

for every pair $(g, d) \in G \times D$. The double integral is strictly positive. □

3. Necessity

It remains to prove that a bad finite normal generates sets G and D satisfying Lemma 4. The proof is slicewise.

For $s \in \mathbf{R}$, define the horizontal boundary of A by

$$\beta(s) := \sup\{x_1 : (x_1, s) \in A\} \in [-\infty, +\infty] \ .$$

Since A is closed, convex, and coordinatewise lower,

$$A \cap \{x_2 = s\} = \{(x_1, s) : x_1 \le \beta(s)\} \ ,$$

with the usual interpretation when $\beta(s) = \pm\infty$. The function β is concave in the extended-real sense. Set

$$m(s) := \beta(s) + s \ , \qquad M(s) := \sup_{r \le s} m(r) \ .$$

Thus $m(s)$ is the largest value of $T_1 = x_1 + x_2$ on the horizontal slice of A, and $M(s)$ is its running maximum from below.

Lemma 5 (Slice formula for $\widehat{A}$)**.** *For every $s \in \mathbf{R}$,*

$$\widehat{A} \cap \{x_2 = s\} = \{(x_1, s) : x_1 \le M(s) - s\} \ .$$

Equivalently,

$$(x_1, s) \in \widehat{A} \quad \Longleftrightarrow \quad x_1 + s \le M(s) \ .$$

Proof. First suppose $(x_1, s) \in A - \Pi$; closure will be handled at the end. Then for some $(y_1, r) \in A$ and some $\eta, \tau \ge 0$,

$$(x_1, s) = (y_1, r) - \eta(1, 0) - \tau(1, -1) \ .$$

Hence $r = s - \tau \le s$ and

$$x_1 + s = y_1 + r - \eta \le m(r) \le M(s) \ .$$

Therefore every point of $A - \Pi$ satisfies $x_1 + s \le M(s)$, and the same is true after taking closures.

Conversely, suppose $x_1 + s < M(s)$. Then there is $r \le s$ such that $m(r) > x_1 + s$. Choose $y_1 \le \beta(r)$ with

$$y_1 + r > x_1 + s;$$

this is immediate if $\beta(r)$ is finite, and trivial if $\beta(r) = +\infty$. Then $(y_1, r) \in A$. Put

$$\tau = s - r \ge 0 \ , \qquad \eta = y_1 + r - x_1 - s > 0 \ .$$

A direct calculation gives

$$(x_1, s) = (y_1, r) - \eta(1, 0) - \tau(1, -1) \in A - \Pi \ .$$

Thus all points with $x_1 + s < M(s)$ belong to $A - \Pi$. Taking closures gives the stated formula with weak inequality. □

Lemma 6 (Gain-loss separation)**.** *Let A' be defined as in Lemma 3, and define the gain and loss sets*

$$G := A \setminus A' = A \cap \{T_1 > b\} \ , \qquad D := A' \setminus A \ .$$

Then

$$P_0(G) = P_0(D) > 0 \ .$$

Moreover, for every $g \in G$ and every $d \in D$,

$$T_1(g) > T_1(d) \qquad \textit{and} \qquad T_2(g) \geq T_2(d) \ .$$

Proof. Since $A \subseteq \widehat{A}$ and $A' = \widehat{A} \cap \{T_1 \leq b\}$,

$$A \setminus A' = A \cap \{T_1 > b\} = G \ .$$

Also $D = A' \setminus A$. Because $P_0(A) = P_0(A') = 1 - \alpha$,

$$P_0(G) = P_0(D) \ .$$

This common mass is positive. Indeed, if it were zero, then A and A' would be equal up to a Gaussian null set. Since both are closed and convex, this would force $A = A'$. But Lemma 3 shows that all finite-support directions of A' lie in $\mathcal{F}$, whereas A has a finite-support direction outside $\mathcal{F}$, a contradiction.

We now prove the ordering. On the slice $x_2 = s$, the gain set is

$$G_s = \{x_1 : (x_1, s) \in G\} = (b - s, \beta(s)] \ .$$

Thus

$$G_s \neq \varnothing \quad \Longleftrightarrow \quad m(s) > b \ . \tag{3}$$

Using Lemma 5, the loss slice is

$$D_s = \{x_1 : (x_1, s) \in D\} = (\beta(s), \min\{M(s), b\} - s] \ .$$

Consequently

$$D_s \neq \varnothing \quad \Longleftrightarrow \quad M(s) > m(s) \ \text{ and } \ m(s) < b \ . \tag{4}$$

Let

$$s_+ := \sup\{s : m(s) > b\} \ .$$

The set in braces is nonempty because $P_0(G) > 0$. It is bounded above because $P_0(D) > 0$ and, as shown next, no loss slice can lie strictly below s_+; if $s_+ = +\infty$, there would be no loss slice at all.

We claim that

$$G_s \neq \varnothing \quad \Longrightarrow \quad s \leq s_+ \ ,$$

and

$$D_s \neq \varnothing \quad \Longrightarrow \quad s \geq s_+ \ .$$

The first implication follows directly from the definition of s_+ and (3). For the second, suppose toward a contradiction that $D_s \neq \varnothing$ for some $s < s_+$. By (4), $M(s) > m(s)$, so there is $q \leq s$ such that

$$m(q) > m(s) \ .$$

Since $s < s_+$, there is $r > s$ such that

$$m(r) > b > m(s) \ .$$

Because $q < s < r$ and m is concave, concavity gives

$$m(s) \geq \lambda m(q) + (1-\lambda) m(r) > m(s)$$

for the appropriate $\lambda \in (0,1)$ satisfying $s = \lambda q + (1-\lambda) r$. This contradiction proves that no loss slice can occur below s_+.

Therefore every $g \in G$ and $d \in D$ satisfy

$$g_2 \leq s_+ \leq d_2 \ ,$$

and hence

$$T_2(g) = -g_2 \geq -d_2 = T_2(d) \ .$$

Also, by definition of G and A',

$$T_1(g) > b \geq T_1(d) \ .$$

This proves the asserted pairwise ordering. □

Lemma 7 (Necessity of feasible normals)**.** *If A has a finite-support direction a with $a_2 > a_1$, then ϕ_A is inadmissible. More precisely, the exact-size test $\phi_{A'}$ constructed above satisfies*

$$P_\mu(X \notin A') > P_\mu(X \notin A) \qquad \textit{for every } \mu \in \Lambda_1 \ .$$

Proof. By Lemma 3, $\phi_{A'}$ is a level-α test. The difference in power is

$$P_\mu(X \notin A') - P_\mu(X \notin A) = P_\mu(G) - P_\mu(D) \ ,$$

where $G = A \setminus A'$ is newly rejected by $\phi_{A'}$ and $D = A' \setminus A$ is newly accepted by $\phi_{A'}$. Lemma 6 verifies the hypotheses of the transfer criterion, Lemma 4. Hence

$$P_\mu(G) > P_\mu(D) \qquad \text{for every } \mu \in \Lambda_1 \ .$$

Thus $\phi_{A'}$ strictly dominates ϕ_A on the restricted alternative. □

Proof of Theorem 1. If all finite normals of A lie in $\mathcal{F}$, Lemma 1 gives α-admissibility against all level-α tests.

Conversely, suppose that the finite-normal condition fails. Since A is coordinatewise lower, any finite-support direction must be nonnegative. Hence failure of the condition means that there is some $a \neq 0$ such that

$$h_A(a) < \infty \ , \qquad 0 \leq a_1 < a_2 \ .$$

Lemma 7 constructs a level-α test that strictly dominates ϕ_A at every $\mu \in \Lambda_1$. Therefore ϕ_A is inadmissible. □

4. Interpretation and examples

The theorem says that, in the restricted problem $\mu_2 \le \mu_1$, the relevant issue is not whether a test uses X_2. The relevant issue is whether it uses X_2 through a face whose normal points in a feasible direction. The feasible normal cone is

$$\mathcal{F} = \mathrm{cone}\{(1,0),(1,1)\} \ .$$

Thus the short test qualifies because its only finite normal is $(1,0)$. The gated-sum test

$$A_{c,b} = \{x_1 \le c,\ x_1 + x_2 \le b\} \ , \qquad \phi_{c,b}(x) = I\{x_1 > c \text{ or } x_1 + x_2 > b\} \ ,$$

qualifies because its finite normals are $(1,0)$ and $(1,1)$, and it genuinely uses X_2. By contrast, the long rectangle

$$A_\ell = \{x_1 \le c_\ell,\ x_2 \le c_\ell\}$$

has the vertical normal $(0,1) \notin \mathcal{F}$ and is therefore inadmissible in the unrestricted class of level-α tests. The construction above replaces the vertical face by a diagonal face, which is exactly the gated-sum improvement.

For a finite rectangle

$$A = \{x_1 \le c_1,\ x_2 \le c_2\} \ ,$$

the slice boundary is $\beta(s) = c_1$ for $s \le c_2$ and $\beta(s) = -\infty$ for $s > c_2$. Hence

$$m(s) = c_1 + s \quad (s \le c_2) \ ,$$

and the separation point in Lemma 6 is $s_+ = c_2$. The competitor is

$$A' = \{x_1 \le c_1,\ x_1 + x_2 \le b\} \ ,$$

with b chosen so that $P_0(A') = P_0(A)$. This is the gated test.

Remark 1 (Scope of the theorem). The characterization is for closed convex coordinatewise lower acceptance regions in the bivariate identity-covariance experiment. The sufficient direction is global, because it uses the Lehmann–Romano ray condition and therefore allows arbitrary randomized level-α competitors. The necessary direction is also global, because the constructed dominator is itself a nonrandomized level-α test. Extensions to $k > 2$ should replace $\mathcal{F}$ by the cone generated by prefix-sum normals, and extensions to $\Sigma \ne I_2$ must express feasibility in natural-parameter directions, not by naively whitening the coordinates.

References

[1] Lehmann, E. L. and Romano, J. P. (2005). *Testing Statistical Hypotheses*. Third edition. Springer.